Guidelines

Oral IRON for patients undergoing pre-operative optimisation of iron stores

Reference #: MM013

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Patient Blood Management Clinical Nurse Consultant	Nursing

Introduction

In line with the National Blood Authority's (NBA) Patient Blood Management (PBM) perioperative guidelines, patients that are waitlisted for surgery where substantial blood loss is expected are screened for anaemia and iron stores are assessed prior to surgery.

When patients are identified with stored iron levels (ferritin) lower than those outlined in the SCGH PBM Treatment algorithm, a recommendation will be made to commence oral iron with the aim of repleting levels prior to surgery.

Although oral iron is available over the counter, there are many preparations that will not be effective. In addition, oral iron can reduce the efficacy of some commonly prescribed medications. Advice regarding suitable oral iron preparations and timing to avoid medication interaction must be provided to minimise potential harm to patients.

1.0 Indications

Pre-operative surgical patients waitlisted for surgery where with sub-optimal stores for surgery with expected significant blood loss AND time to surgery > 6 weeks (IV iron recommended if <6 weeks until surgery):

- With iron deficiency +/- anaemia
- With sub-optimal iron stores for surgery (as defined by ferritin <100 ug/L)

2.0 Oral Iron Dosing

The usual ADULT dose for Iron Deficiency Anaemia (IDA) is around 100-200 mg elemental iron daily in divided doses (1-2 tablets of these preparations per day, ideally 1 hr before or 2 hrs after food).
Recommended therapeutic oral iron preparations in Australia

In adults, lower doses can be administered as either intermediate dose tablets (containing 30-60 mg of elemental iron) or intermittent dosing (e.g. second daily to weekly). Either approach may be useful in patients with mild IDA and a long lead in time until surgery.

- Patients with IDA may need oral iron for 3-6 months to replete iron stores. GI upset may be reduced by taking iron tablets with food, or at night and increasing the dose gradually.

- Many oral iron preparations contain too little iron to be effective. Multivitamin-mineral supplements should not be used to treat IDA as the iron content is low.
- If a prescription is required, the patient may be referred to their GP.

2.1 Recommended Therapeutic Oral Iron Tablets

Medicine	Dose Range	Hyperlink
Ferro-f-tab®	**1-2 tablets daily**	**See MIMS online**
Ferro-Grad C®	**1-2 tablets daily**	**See MIMS online**
Ferro-Grad®	**1-2 tablets daily**	**See MIMs online**

3.0 Precautions

The **combination** of oral iron and some prescribed medications can have deleterious effects.

- It is important to check currently prescribed medications before commencing oral iron.
- Advise the patient to check with the pharmacist for advice regarding drug interactions.
- Refer to MIMS online for specific drug interactions.

Below is a list of **SOME** commonly prescribed medicines that interact with oral iron.

Condition	Medicines (examples)	Adverse Effect	Implication
Parkinson's Disease	Levodopa and Carbidopa	Iron chelates causing significant reduction in the absorption of these drugs	Separate dosing as far as possible. Consider IV iron.
Antimicrobials, infection, acne	Tetracycline Minocycline Doxycycline Ciprofloxacin Moxifloxacin Norfloxacin	Iron chelates causing significant reduction in the absorption of these drugs	Separate administration by at least 3 hours
Many indications: e.g. Rheumatoid Arthritis, Wilson's disease, lead poisoning	Penicillamine	Iron chelates causing significant reduction in the absorption of this drug	Separate administration by at least 2 hours
Thyroid Hormone Deficiency	Levothyroxine (thyroxine) Liothyronine	Oral iron salts may reduce the absorption and effect of thyroid hormone by forming poorly soluble complex with thyroid hormone	Separate administration by at least 2 hours
Epilepsy	Phenytoin combined with iron tablets that contain folic acid (Ferro-f-tab®)	The use of folic acid may increase the hepatic metabolism of the anticonvulsant, resulting in decreased serum levels and effect of the anticonvulsant. Concurrent use of folic or folinic acid and phenytoin has led to loss of seizure control.	Avoid Ferro-f-tab®
Immunosuppressant	Mycophenolate	Reduced absorption of	Avoid

		mycophenolate (up to 90%)	combination. Separate by a minimum of 4 hrs.
Osteoporosis	Bisphosphonates (alendronate, risedronate)	Bisphosphonates form complexes with polyvalent cations e.g. iron. This results in reduced absorption of oral bisphosphonate	Separate administration by at least 2 hours
Osteoporosis	Calcium	Iron absorption reduced	Avoid combination. Separate by several hours.
GORD, dyspepsia, reflux	Antacids, PPIs	Iron absorption reduced	Avoid combination. Separate by as long as possible.

4.0 Iron deficiency Anaemia: GP notification

When a patient is identified with iron deficiency (ferritin < 30 microg/L), a letter should be sent to the patient requesting them to follow up with their GP within 2 weeks, and to their GP recommending investigation of the iron deficiency. Ensure that a copy of relevant blood results is enclosed.

Related Documents

Legislation:

- Medicines and Poisons Act 2014
- Medicines and Poisons Regulations 2016
- Health Services Act 2016

Authorising Policy and Standard/s:

- SCGH Hospital Policy 141 Medicines Management
- Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy
- Nursing Practice Guideline No.51 Medication Management
- NSQHS Standard 1.1.1: An organisation-wide management system is in place for the development, implementation and regular review of policies, procedures and/or protocols.
- NSQHS Standard 1.7: Developing and/or applying clinical guidelines or pathways that supported by best available evidence (all subsequent actions of this criterion are applicable)
- NSQHS Standard 4.1: Developing and implementing governance arrangements and organisational policies, procedures and/or protocols for medication safety, which are consistent with national and jurisdictional legislative requirements, policies and guidelines.
- NSQHS Standard 4.1.2: Policies, procedures and/or protocols are in place that are consistent with legislative, national, jurisdictional and professional guidelines.

References

1. National Blood Authority. Patient Blood Management Guidelines: Module 2-Perioperative. (2012). Available from: http://www.blood.gov.au/pbm-guidelines

2. MIMS Online. Australia. 2020. Available from: https://www-mimsonline-com-au.qelibresources.health.wa.gov.au/Search/Search.aspx
3. Department of Health Western Australia. Patient Blood management. Oral preparations for treatment of iron deficiency anaemia in Australia. (2011). Available from: https://ww2.health.wa.gov.au/~/media/Files/Corporate/general%20documents/Patient%20blood%20management/Oral_preparations_treatment_of_iron_deficiency.ashx
4. BloodSafelearning, Patient Blood Management Courses. Iron deficiency anaemia. Available from: https://bloodsafelearning.org.au/
5. Munoz M, Acheson A, Auerbach M, Besser M, Habler O, Kehlet H, et al. International consensus statement of the peri-operative management of anaemia and iron deficiency. Anaesthesia. 2017;72:233-247. Doi:10.111/anae.13773

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